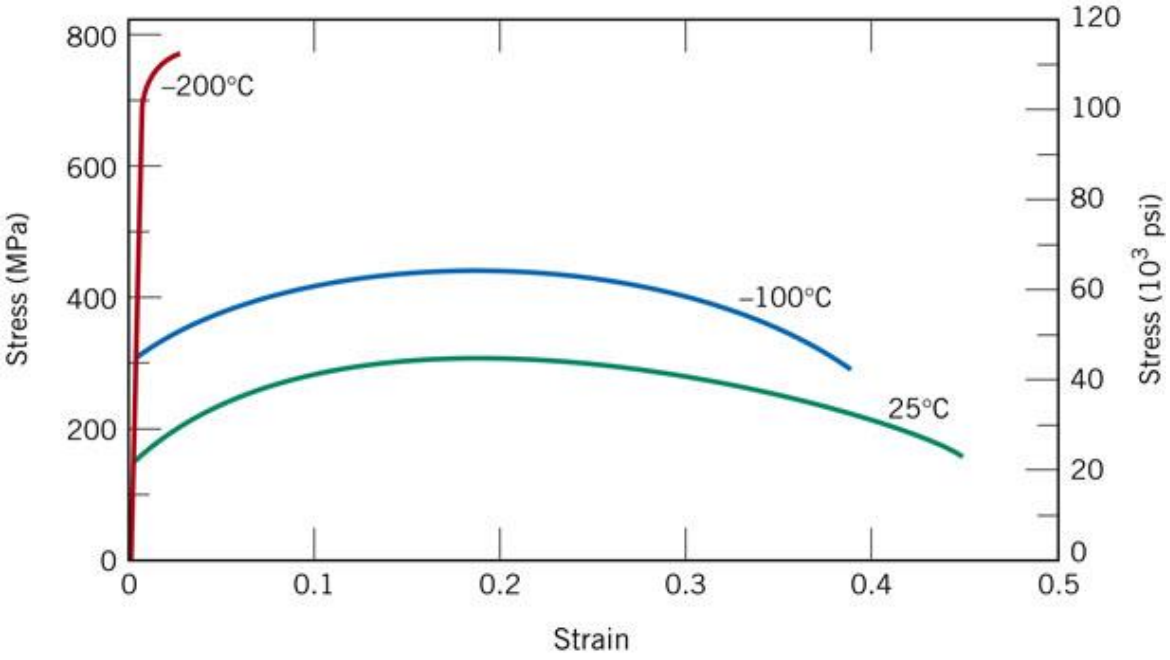
Outline:

- Ductility
- Resilience
- Toughness
- Hardness
- Example
- Slip Systems

Effect of Testing Temperature on Mechanical Behavior



- *The yield and tensile strengths …… with increasing temperature.*
- *Ductility …… with temperature.*

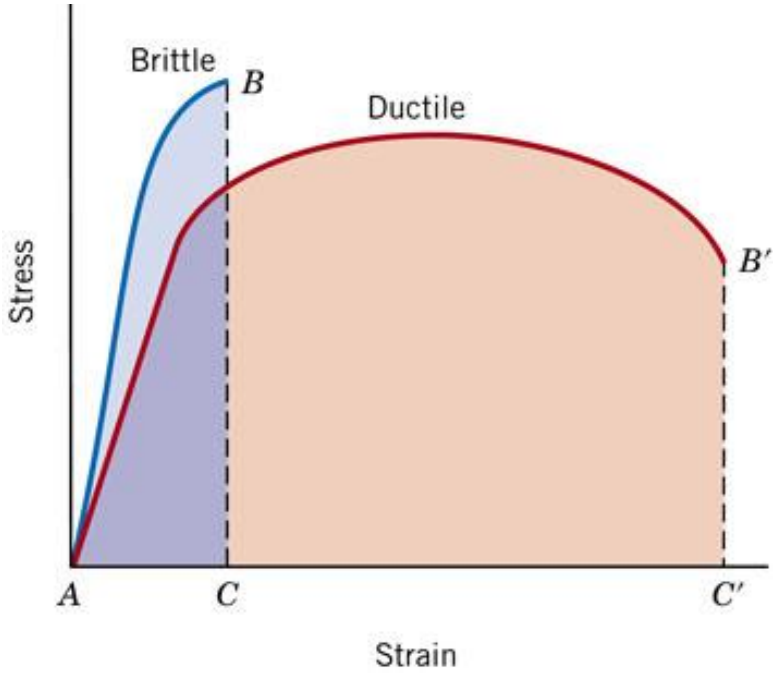
DUCTILITY

Ductility measures the amount of plastic deformation that a material goes through by the time it breaks.

Two measures of ductility:

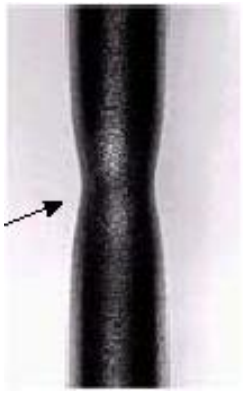
1) Percent Elongation **(%*El*)**

$$\%\,\mathrm{El} = \frac{\text{Final } \textit{fractured} \text{ length - Initial length}}{\text{Initial length}} \times 100$$



2) Percent Reduction In Area

$$\%\mathrm{RA} = \frac{\text{Initial Area - Final Area}}{\text{Initial Area}} \times 100$$



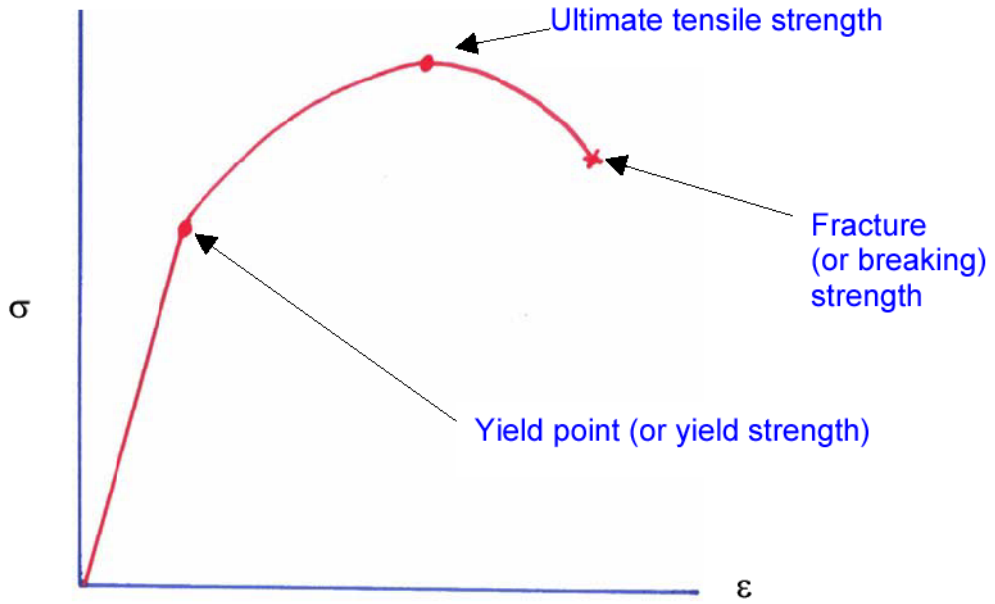
DUCTILITY

- Ductility is a measure of how much strain a given stress produces.
- Highly *ductile* metals can exhibit significant strain before fracturing, whereas *brittle* materials frequently display very little strain.
- An overly simplistic way of viewing ductility is the degree to which a material is "forgiving" of local deformation without the occurrence of fracture.

Brittle materials: %EL ≤ 5% at fracture

Ductile materials: %EL and %RA both ≥ 25%

Typical Mechanical Properties of Metals



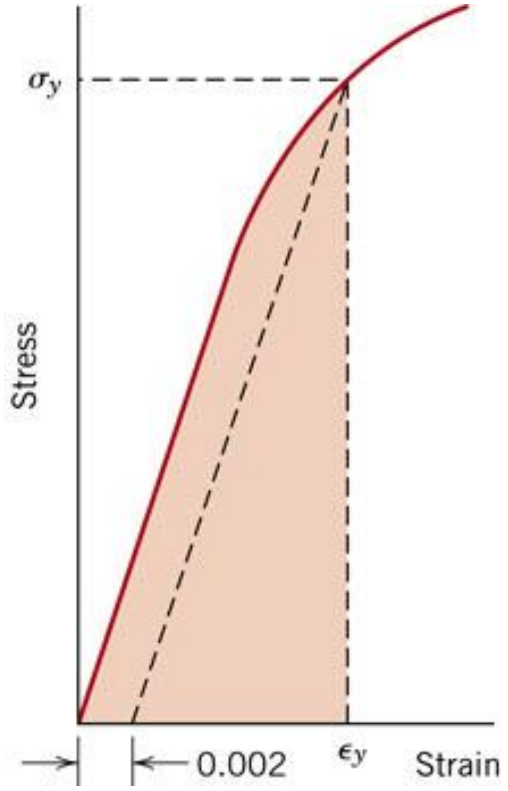
Metal Alloy	*Yield Strength MPa (ksi)*	*Tensile Strength MPa (ksi)*	*Ductility, %EL [in 50 mm (2 in.)]*
Aluminum	35 (5)	90 (13)	40
Copper	69 (10)	200 (29)	45
Brass (70Cu–30Zn)	75 (11)	300 (44)	68
Iron	130 (19)	262 (38)	45
Nickel	138 (20)	480 (70)	40
Steel (1020)	180 (26)	380 (55)	25
Titanium	450 (65)	520 (75)	25
Molybdenum	565 (82)	655 (95)	35

RESILIENCE

Ability of material to absorb energy during **elastic** deformation and then to **give it back** when unloaded.

- Measured with **Modulus of Resilience**, $\mathbf{U_r}$
- $\mathbf{U_r}$, is area under **σ - ε** curve up to yielding:

$$U_r = \int_0^{\varepsilon_y} \sigma \, d\varepsilon$$



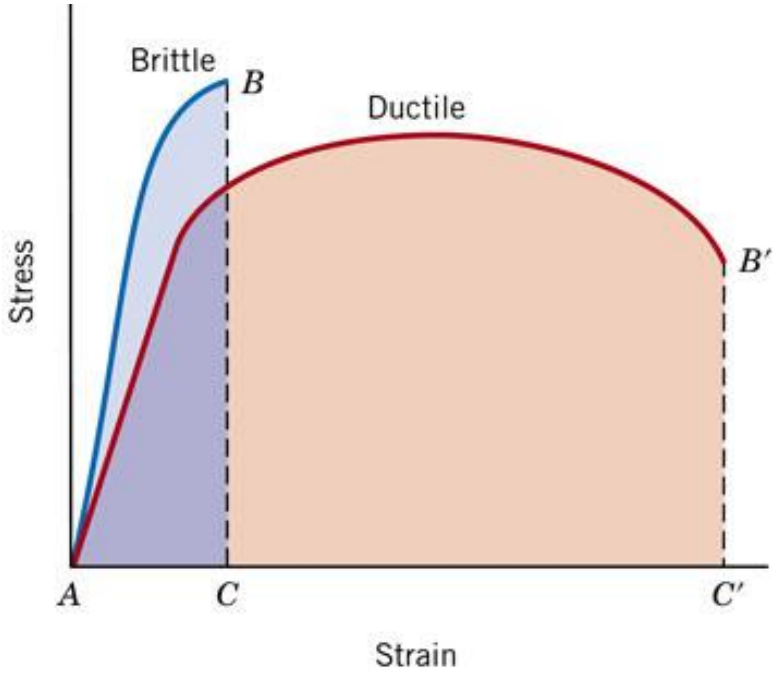
- Assuming a **linear elastic** region:

$$U_r = \frac{1}{2}\sigma_y \varepsilon_y = \frac{1}{2}\sigma_y \left(\frac{\sigma_y}{E}\right) = \frac{\sigma_y^2}{2E}$$

- Units are J/m^3 (equivalent to ……)

TOUGHNESS

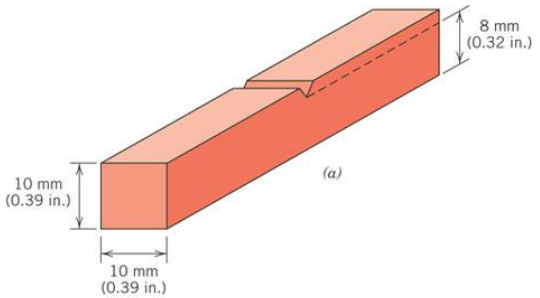
Ability to absorb energy before **fracture**



- Toughness is the **area under σ - ε curve up to fracture**.
  - *Similar to Resilience (same units J/m³).*
  - *Larger area ⇒ ..... material.*
- So tough materials have a **combination of strength and ductility**
- Can be measured by an impact test *(Chapter 8)*.

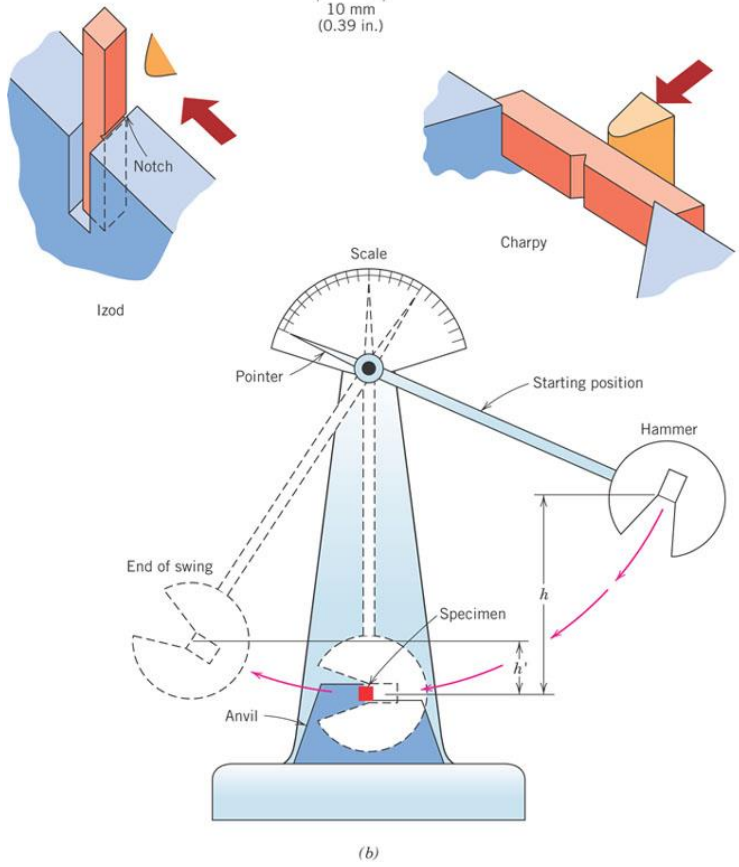
Toughness Measurement: *Impact Testing*

- *Charpy and Izod* tests measure *impact energy or notch toughness*
- Charpy V- notch (CVN) *most common*



Charpy Test

- use *standard sized bar* specimens with a central notch
- weighted pendulum released from a height, h
- impacts the specimen behind the notch (*stress concentration*)
- fracture of specimen occurs and energy is absorbed
- the pendulum travels to point, h´, where h´< h
- obtain the amount of absorbed energy from scale



HARDNESS

Hardness is a measure of the material's resistance to localized plastic deformation (e.g. dent or scratch)

Qualitative Hardness:

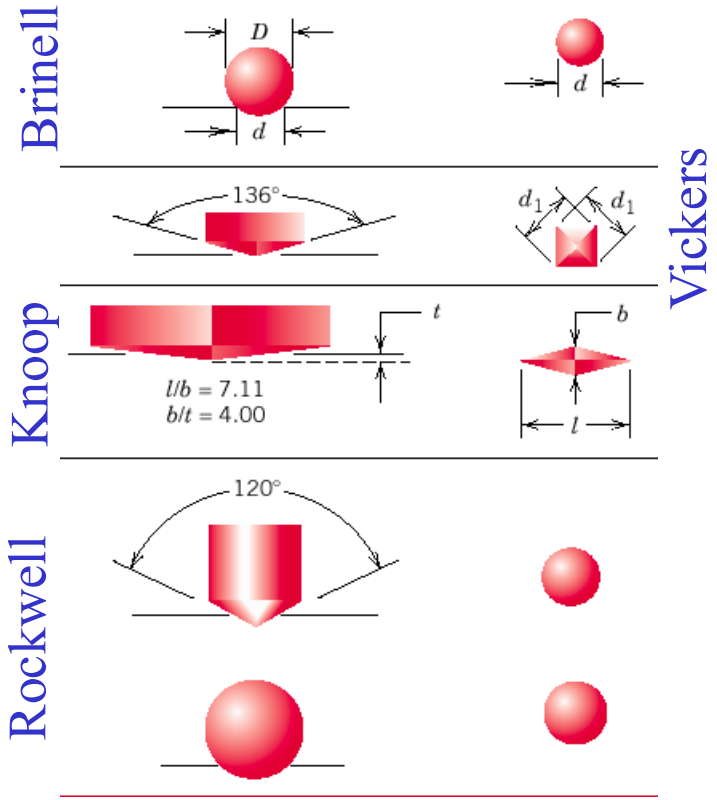
- Moh's scale, determined by the ability of a material to scratch another material:

from 1 (softest = talc) to 10 (hardest = diamond)

Quantitative Hardness:

- Different types of quantitative hardness test have been designed
  - Rockwell
  - Brinell
  - Vickers
  - Knoop

HARDNESS



- Usually a small indenter (*sphere, cone, or pyramid*) is forced into the surface of a material under conditions of controlled magnitude and rate of loading.
- The depth or size of indentation is measured.
- The tests are somewhat approximate, but popular because they are easy and non-destructive (except for the small dent).

$$\mathrm{HB} = \frac{2P}{\pi D[D - \sqrt{D^2 - d^2}]}$$

Where,
P (the applied load) is in kg,
D is the indenter's diameter
d is the diameter of the resulted indentation

Example

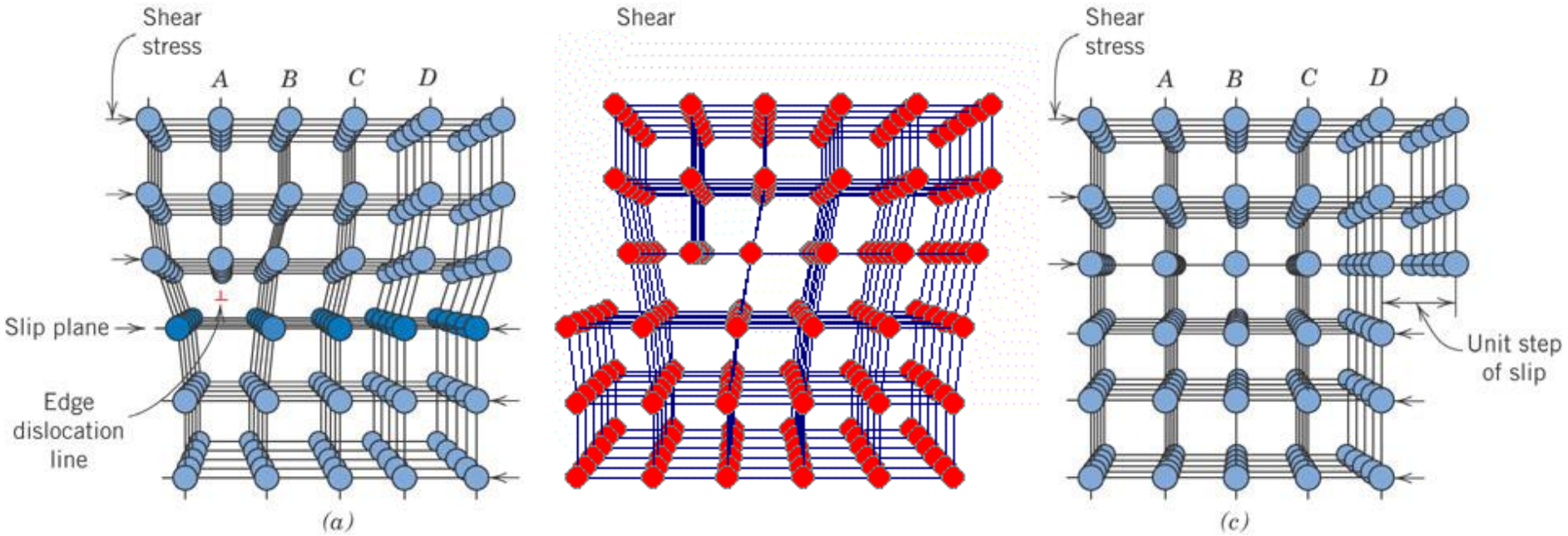
A cylindrical metal specimen having an original diameter of 12.8 mm (0.505 in.) and gauge length of 50.80 mm (2.000 in.) is pulled in tension until fracture occurs. The diameter at the point of fracture is 6.60 mm (0.260 in.), and the fractured gauge length is 72.14 mm (2.840 in.). Calculate the ductility in terms of percent reduction in area and percent elongation.

Plastic Deformation

- Why metals could be plastically deformed?
- Why the plastic deformation properties could be changed to a very large degree by forging without changing the chemical composition?
- Why plastic deformation occurs at stresses that are much smaller than the theoretical strength of perfect crystals?
- The force to break all bonds in the slip plane is much higher than the force needed to cause the deformation. Why?

These questions can be answered based on the idea proposed in 1934 by Taylor, Orowan and Polyani: ***Plastic deformation is due to the motion of a large number of ……..***

Dislocations allow deformation at much lower stress than in a perfect crystal, How?!



The movement of the dislocation (to the right in this sequence) requires the breaking (and formation) of only ONE set of bonds per step.

Dislocations move in ..... directions within ..... planes.

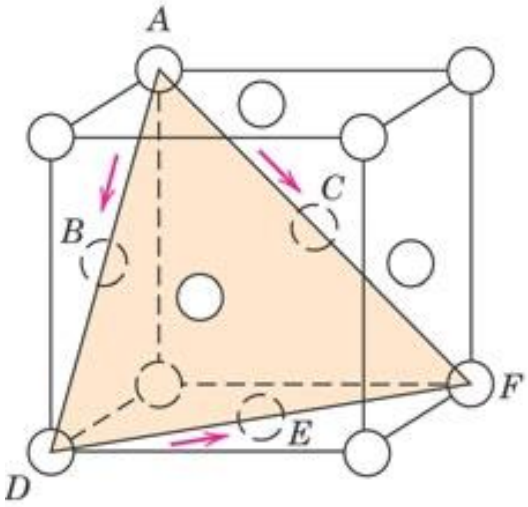
Dislocations and Plastic Deformation

Under applied shear stress, dislocations can move by breaking bonds **CONSECUTIVELY** (rather than simultaneously) ➔Requires less energy.

This is the reason why experimental shear strength is lower.

Deformation by dislocations movement is called **SLIP.**

• The combination of C-P plane (the slip plane) and C-P direction (the slip direction) is called a ……..



Recall:

SLIP SYSTEMS DEPEND ON THE CRYSTAL STRUCTURE OF THE MATERIAL!

Dislocations and Plastic Deformation

Table 7.1 Slip Systems for Face-Centered Cubic, Body-Centered Cubic, and Hexagonal Close-Packed Metals

Metals	*Slip Plane*	*Slip Direction*	*Number of Slip Systems*
	Face-Centered Cubic		
Cu, Al, Ni, Ag, Au	{111}	$\langle 1\bar{1}0 \rangle$	12
	Body-Centered Cubic		
α-Fe, W, Mo	{110}	$\langle \bar{1}11 \rangle$	12
α-Fe,W	{211}	$\langle \bar{1}11 \rangle$	12
α-Fe, K	{321}	$\langle \bar{1}11 \rangle$	24
	Hexagonal Close-Packed		
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	$\{10\bar{1}0\}$	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	$\{10\bar{1}1\}$	$\langle 11\bar{2}0 \rangle$	6

The more slip systems available, the easier it is for dislocations to move, which is why (on the average) FCC and BCC metals are more ductile than HCP metals.

Next Topic:

Phase Diagrams

The midterm exam is on Friday October 30th